

CSE-4E3: Design Data Intensive Applications

LAB1: OLAP

Team members

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Deliverables :

1. Steps showing generation of TPC data

1. from the directory you want to generate data run this command to download tpch-dbgen source code :

“git clone https://github.com/electrum/tpch-dbgen.git”

```
nayra@nayra-VirtualBox:~$ cd "/Design Data Intensive Applications/Lab_1_OLAP"
nayra@nayra-VirtualBox:~/Design Data Intensive Applications/Lab_1_OLAP$ git clone https://github.com/electrum/tpch-dbgen.git
```

2. In the downloaded directory (tpch-dbgen), edit the file `makefile.suite` and set the following variables {CC , DATABASE , MACHINE , WORKLOAD} to the values specified.

```
99 #
100 #####
101 ## CHANGE NAME OF ANSI COMPILER HERE
102 #####
103 CC      = gcc
104 # Current values for DATABASE are: INFORMIX, DB2, TDAT (Teradata)
105 #                                     SQLSERVER, SYBASE, ORACLE
106 # Current values for MACHINE are:  ATT, DOS, HP, IBM, ICL, MVS,
107 #                                     SGI, SUN, U2200, VMS, LINUX, WIN32
108 # Current values for WORKLOAD are:  TPC
109 DATABASE= INFORMIX
110 MACHINE  = LINUX
111 WORKLOAD = TPC
```

3. in the downloaded directory run this command to make utility :

“make -f makefile.suite”

```
nayra@nayra-VirtualBox:~/Design Data Intensive Applications/Lab_1_OLAP$
nayra@nayra-VirtualBox:~/Design Data Intensive Applications/Lab_1_OLAP$ cd ./tpch-dbgen
nayra@nayra-VirtualBox:~/Design Data Intensive Applications/Lab_1_OLAP/tpch-dbgen$ make -f makefile.suite
```

- if make or gcc commands aren't recognized by system run these commands :

“sudo apt install make”

“sudo apt install gcc”

- create new directory to generate data files in it and copy the dbgen executable file and dists.dss file from tpch-dbgen directory to it. then from this new directory run this command to generate data files:

“./dbgen -s 1”

```
nayra@nayra-VirtualBox:~/Design Data Intensive Applications/Lab_1_OLAP/Data$  
nayra@nayra-VirtualBox:~/Design Data Intensive Applications/Lab_1_OLAP/Data$ ./dbgen -s 1
```

-s 1 : refers to scale factor = 1 , scale factor = 1 means that data will be of size 1GB .

- run this command to check if data has been generated successfully.

“ls -l”

NOW Data is generated successfully .

```
nayra@nayra-VirtualBox:~/Design Data Intensive Applications/Lab_1_OLAP/Data$ ls -l  
total 1075068  
-rw-rw-r-- 1 nayra nayra 24346144 Feb 21 19:18 customer.tbl  
-rwxrwxr-x 1 nayra nayra 135608 Feb 21 19:17 dbgen  
-rw-rw-r-- 1 nayra nayra 11815 Feb 21 19:18 dists.dss  
-rw-rw-r-- 1 nayra nayra 759863287 Feb 21 19:18 lineitem.tbl  
-rw-rw-r-- 1 nayra nayra 2224 Feb 21 19:18 nation.tbl  
-rw-rw-r-- 1 nayra nayra 171952161 Feb 21 19:18 orders.tbl  
-rw-rw-r-- 1 nayra nayra 118984616 Feb 21 19:18 partsupp.tbl  
-rw-rw-r-- 1 nayra nayra 24135125 Feb 21 19:18 part.tbl  
-rw-rw-r-- 1 nayra nayra 389 Feb 21 19:18 region.tbl  
-rw-rw-r-- 1 nayra nayra 1409184 Feb 21 19:18 supplier.tbl  
nayra@nayra-VirtualBox:~/Design Data Intensive Applications/Lab_1_OLAP/Data$
```

2. Steps showing insertion of generated data into MySQL.

1. the value of secure_file_priv means that only files from this directory could be loaded into mysql.

```
mysql>
mysql>
mysql> SHOW VARIABLES LIKE 'secure_file_priv' ;
+-----+-----+
| Variable_name | Value                               |
+-----+-----+
| secure_file_priv | /var/lib/mysql-files/             |
+-----+-----+
1 row in set (0.73 sec)
```

2. so we will copy data files to this directory.
3. run these commands to get into this directory :

```
“ cd /var/lib
  sudo chmod 777 mysql-files
  cd ./mysql-files”
```

sudo chmod 777 mysql-files -> to give permission for editing mysql-files directory.

```
nayra@nayra-VirtualBox:~/Design Data Intensive Applications/Lab_1_OLAP/Data$ cd ~
nayra@nayra-VirtualBox:~$ cd ..
nayra@nayra-VirtualBox:/home$ cd ..
nayra@nayra-VirtualBox:/home$ cd /var
nayra@nayra-VirtualBox:/var$ cd ./lib
nayra@nayra-VirtualBox:/var/lib$ sudo chmod 777 ./mysql-files
[sudo] password for nayra:
nayra@nayra-VirtualBox:/var/lib$ cd ./mysql-files
nayra@nayra-VirtualBox:/var/lib/mysql-files$
```

4. Now you are in the required directory and you only want to copy or move data files into this directory. You could use this commands :

“cp <path to source file> <destination directory>” for coping.

OR

“mv <path to source file> <destination directory>” for moving.

you could replace <destination directory> with “.” because you are already in the destination directory.

```
nayra@nayra-VirtualBox:/var/lib/mysql-files$ cp "/home/nayra/Design Data Intensive Applications/Lab_1_OLAP/Data/region.tbl" .
nayra@nayra-VirtualBox:/var/lib/mysql-files$ cp "/home/nayra/Design Data Intensive Applications/Lab_1_OLAP/Data/nation.tbl" .
nayra@nayra-VirtualBox:/var/lib/mysql-files$ cp "/home/nayra/Design Data Intensive Applications/Lab_1_OLAP/Data/customer.tbl" .
nayra@nayra-VirtualBox:/var/lib/mysql-files$ cp "/home/nayra/Design Data Intensive Applications/Lab_1_OLAP/Data/lineitem.tbl" .
nayra@nayra-VirtualBox:/var/lib/mysql-files$ cp "/home/nayra/Design Data Intensive Applications/Lab_1_OLAP/Data/orders.tbl" .
nayra@nayra-VirtualBox:/var/lib/mysql-files$ cp "/home/nayra/Design Data Intensive Applications/Lab_1_OLAP/Data/part.tbl" .
nayra@nayra-VirtualBox:/var/lib/mysql-files$ cp "/home/nayra/Design Data Intensive Applications/Lab_1_OLAP/Data/partsupp.tbl" .
nayra@nayra-VirtualBox:/var/lib/mysql-files$ cp "/home/nayra/Design Data Intensive Applications/Lab_1_OLAP/Data/supplier.tbl" .
```

```
nayra@nayra-VirtualBox:/var/lib/mysql-files$ ls -l
total 1074916
-rw-rw-r-- 1 nayra nayra 24346144 Feb 21 20:06 customer.tbl
-rw-rw-r-- 1 nayra nayra 759863287 Feb 21 20:06 lineitem.tbl
-rw-rw-r-- 1 nayra nayra 2224 Feb 21 20:05 nation.tbl
-rw-rw-r-- 1 nayra nayra 171952161 Feb 21 20:07 orders.tbl
-rw-rw-r-- 1 nayra nayra 118984616 Feb 21 20:07 partsupp.tbl
-rw-rw-r-- 1 nayra nayra 24135125 Feb 21 20:07 part.tbl
-rw-rw-r-- 1 nayra nayra 389 Feb 21 20:05 region.tbl
-rw-rw-r-- 1 nayra nayra 1409184 Feb 21 20:08 supplier.tbl
nayra@nayra-VirtualBox:/var/lib/mysql-files$
```

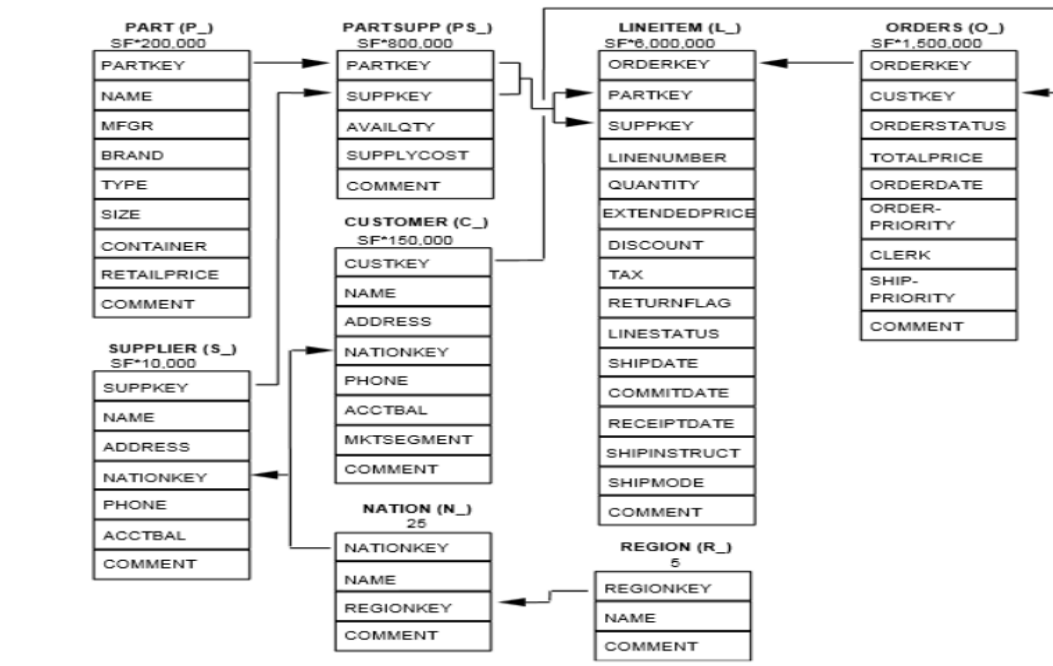
NOW Data files are in the right directory and we will only run sql script to insert data into MySQL.

3. DDL of the OTLP schema in MySQL

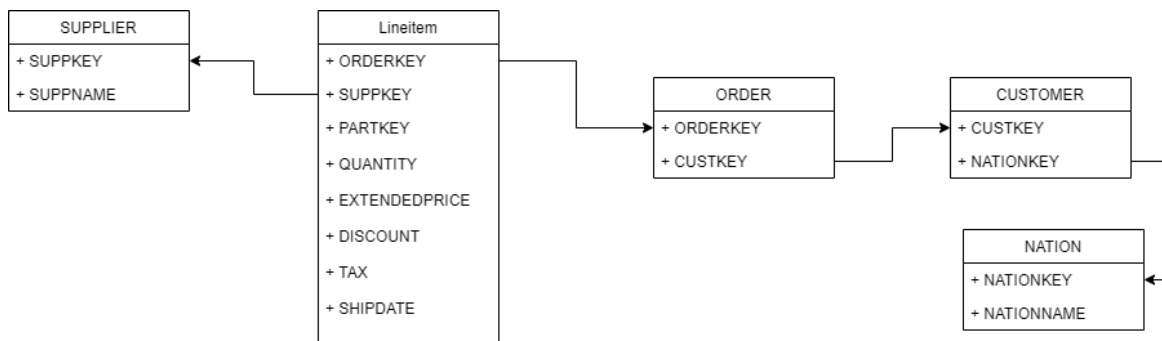
Exist in the Github repo and attached with deliverables.

DataBase design and schemas

Relational Database design



Relations Database Design after applying reduction on tables according to required query



The query that the design is based on

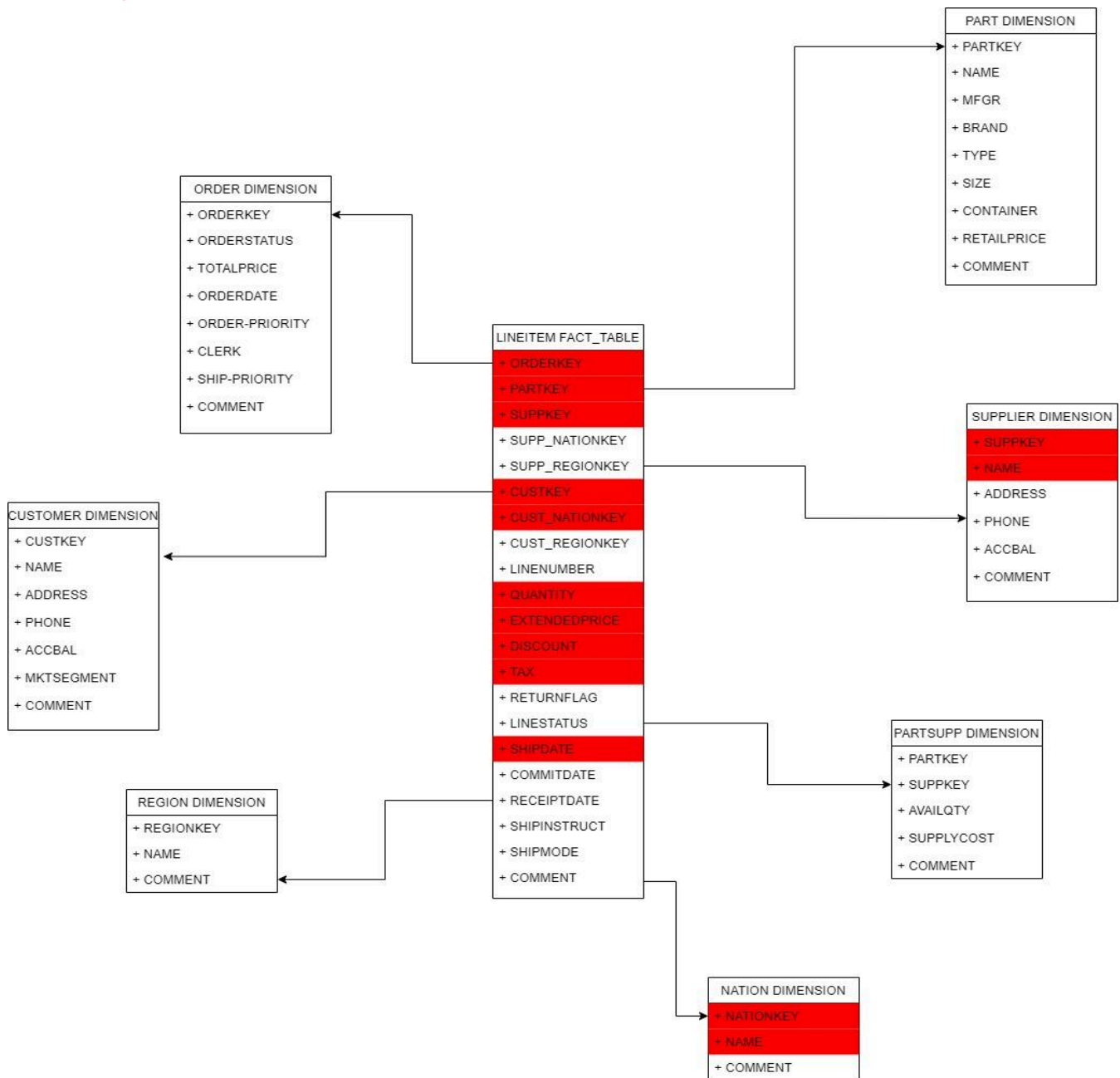
```
205 SELECT
206     n.n_name,
207     s.s_name,
208     SUM(l.l_quantity) AS sum_qty,
209     SUM(l.l_extendedprice) AS sum_base_price,
210     SUM(l.l_extendedprice * (1 - l.l_discount)) AS sum_disc_price,
211     SUM(l.l_extendedprice * (1 - l.l_discount) * (1 + l.l_tax)) AS sum_charge,
212     AVG(l.l_quantity) AS avg_qty,
213     AVG(l.l_extendedprice) AS avg_price,
214     AVG(l.l_discount) AS avg_disc,
215     COUNT(*) AS count_order
216 FROM Lineitem l
217 JOIN Orders o ON l.l_orderkey = o.o_orderkey
218 JOIN Customer c ON o.o_custkey = c.c_custkey
219 JOIN Nation n ON c.c_nationkey = n.n_nationkey
220 JOIN Supplier s ON l.l_suppkey = s.s_suppkey
221 WHERE l.l_shipdate <= '1998-09-02'
222 GROUP BY n.n_name, s.s_name;
223
```

Recommended Steps for relational database reduction

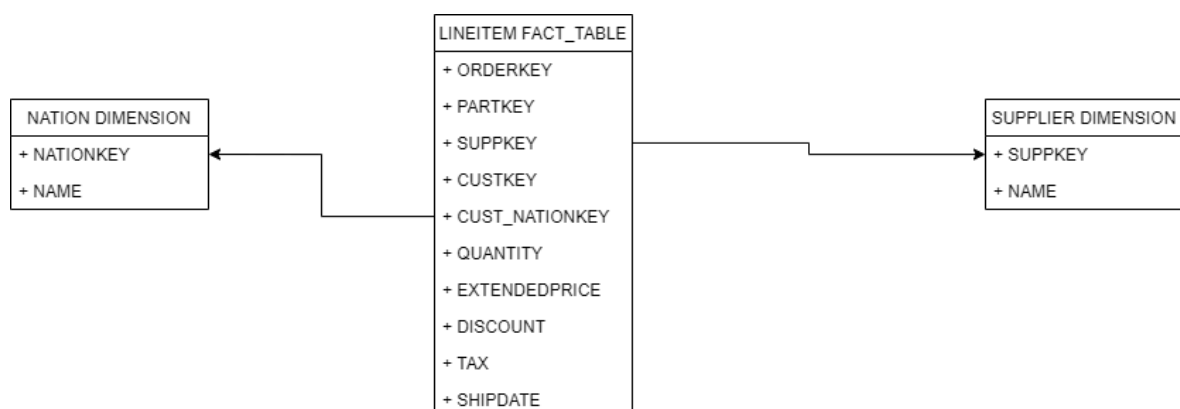
1. extract (Lineitem table , Orders table , Customer table , Nation table , Supplier table) to Nifi.
2. We will not extract (Region table , Partsupp table , part table) because we will not use them.
3. we will remove unnecessary columns from extracted tables to reduce schema.

Star Schema for whole relational database design Based on Lineitem table as the fact table

(Fields filled with the red are fields that we will use in the required query)



Star Schema after reduction



Recommended Steps for reducing star schema from reduced relations database schema

- 1. Fact table <= Lineitem JOIN Orders ON orderkey
JOIN Customer ON Custkey**
- 2. Supplier table & Nation table will be the dimension tables**
- 3. Fact table will reference Nation table with
CUST_NATIONKEY and will reference Supplier table with
SUPPKEY.**

4. DDL of the Star schema or equivalent of transformation queries in NiFi or parquet schemas.

fact table query

```
SELECT
    s.s_suppkey,
    l.l_partkey,
    o.o_orderkey,
    c.c_custkey,
    n.n_nationkey,
    l.l_shipdate,
    l.l_quantity,
    l.l_extendedprice,
    l.l_discount,
    l.l_tax
FROM Lineitem l
JOIN Supplier s ON l.l_suppkey = s.s_suppkey
JOIN Orders o ON l.l_orderkey = o.o_orderkey
JOIN Customer c ON o.o_custkey = c.c_custkey
JOIN Nation n ON c.c_nationkey = n.n_nationkey;
```

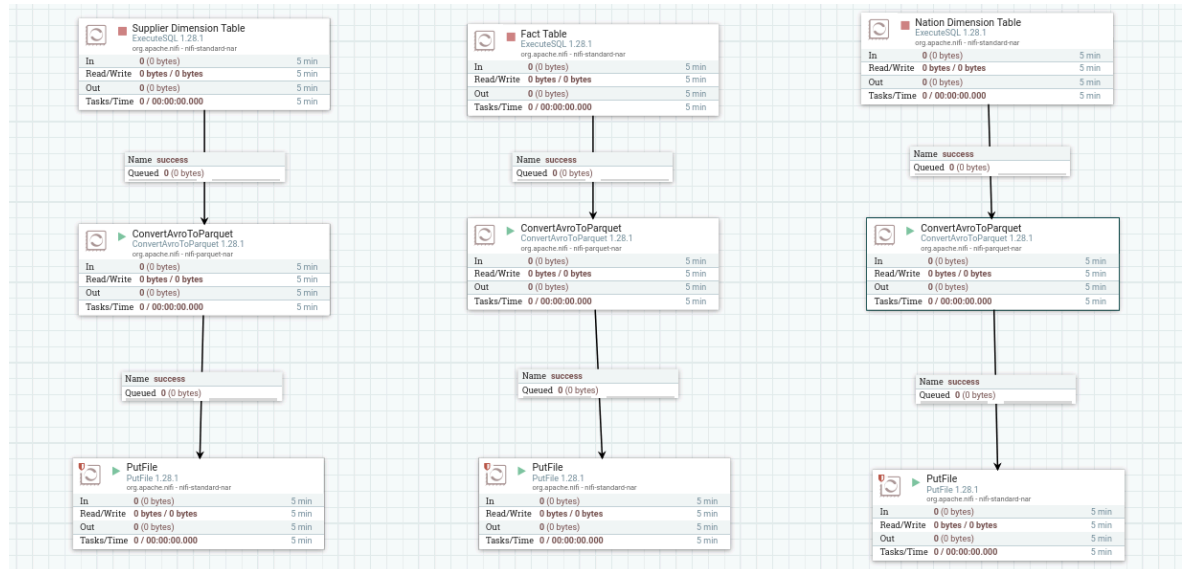
nation table query

```
SELECT
    n.n_nationkey,
    n.n_name
FROM Nation n
JOIN Customer c ON n.n_nationkey = c.c_nationkey;
```

supplier table query

```
SELECT
    s.s_suppkey,
    s.s_name
FROM Supplier s
JOIN Lineitem l ON s.s_suppkey = l.l_suppkey;
```

5. NiFi ETL graph with a brief description of each processor in the processing graph.



NiFi Processors Overview

1. ExecuteSQL

- Executes an SQL query against a database and retrieves results.
- Outputs the data in Avro format for further processing.

2. ConvertAvroToParquet

- Converts Avro data (from ExecuteSQL) into Parquet format.
- Allows specifying compression options and schema settings.

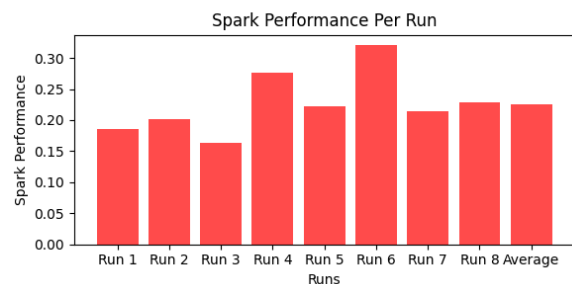
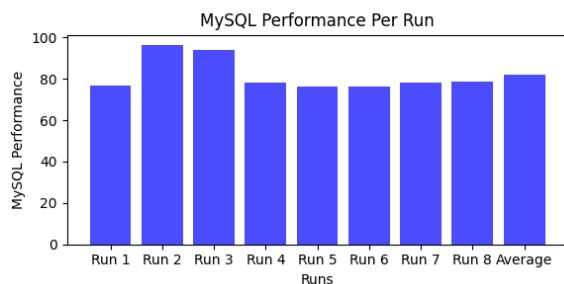
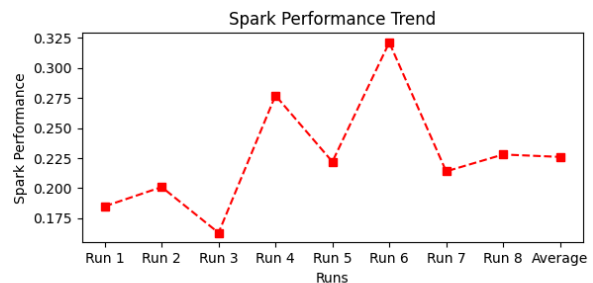
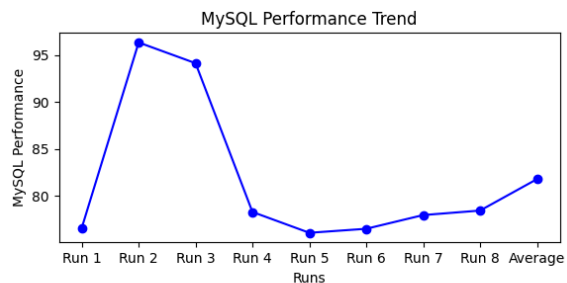
3. PutFile

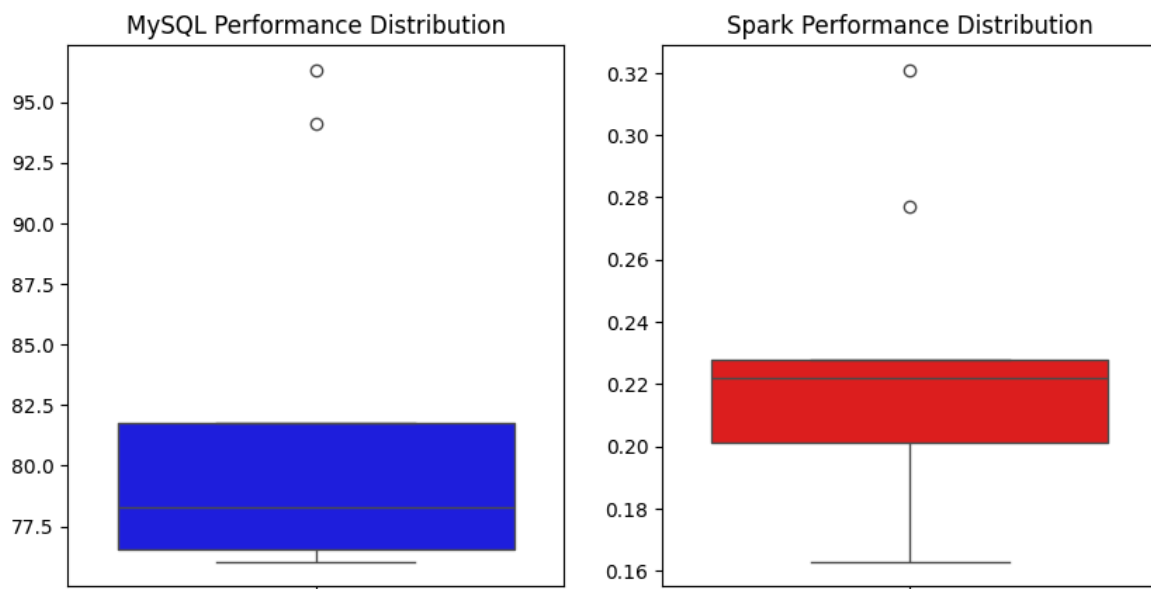
- Saves the Parquet file to a specified directory on the local file system.
- Useful for storing processed data before further use.

6. Table of the 8 runs on MySQL and Spark showing time taken in each of the 16 runs.

	MySQL	Spark
run 1	76.57	0.185
run 2	96.33	0.201
run 3	94.12	0.163
run 4	78.27	0.277
run 5	76.05	0.222
run 6	76.48	0.321
run 7	77.94	0.214
run 8	78.42	0.228
average	81.77	.226

Visualization





GitHub Link.

<https://github.com/mohassan5286/DDIA-/tree/main>